Between the Samples: Spectral Sensing and Operational AI for Industrial Pretreatment

Alex Towfigh, Founder and CEO, AquaMesh Inc.

Session format	50 minute technical session including audience interaction (1 CWEA contact hour)
Topic category	Pretreatment and Industrial Programs; Innovation (emerging technology, AI, digital tools)
Experience level	Intermediate (no spectroscopy background assumed)
Intended audience	POTW pretreatment coordinators and inspectors, industrial user environmental staff, laboratory and engineering staff, consultants, regulators

Abstract

A pretreatment program sees each significant industrial user on a handful of days a year. Between those days, discharges change unobserved. In California alone, recent cases include a refinery that sent more than 300 times its oil and grease limit to the sewer in a few early-morning hours, and a rejected hauler load that came back and dropped a treatment plant's influent pH to 2.4. Composite samples average away the hour that did the damage, and by the time a lab result comes back, the water is long gone.

This session examines what continuous optical sensing and an AI layer can add between the samples, and what they cannot. We explain how ultraviolet-visible absorbance, fluorescence, near-infrared and Raman readings become real-time estimates of organic load, BOD, nitrate, color, oil and solids, and why fouling, drift and changing wastewater chemistry degrade them unless the probe is calibrated against local laboratory results. Attendees read illustrative spectra, set an alarm threshold against a false-alarm budget, and debate where a probe should go first.

We then follow one event from a spike to a case file: detection against each stream's own normal, pointing to the likely discharger, triggering a Part 136 sample while the slug is still passing, and assembling the evidence. We cover what the same AI layer can take off a program's plate, from report drafting and compliance deadlines to staff training and institutional memory, and the guardrails that keep people in charge: validated limits, read-only access by default, approval for every action and a full audit trail. Recent California cases, and programs that already monitor continuously, show what changes when a program sees every day of the year instead of three.

Learning objectives

By the end of the session, attendees will be able to:

1. Identify where scheduled sampling, self-monitoring and inspections leave gaps between samples, and what those gaps cost a municipal program, using recent California cases.
2. Explain how optical probes estimate pretreatment parameters in real time, including BOD, organic load, nitrate, oil and solids, and how fouling, drift and local calibration limit them.
3. Evaluate a continuous-monitoring and AI proposal on validation, data control, guardrails, and enforceability under Part 136.

Presenter

Alex Towfigh is the founder and CEO of AquaMesh Inc., which builds full-spectrum optical water-quality sensors and on-site AI for water and wastewater. He has more than six years of hard-tech startup and R&D experience developing sensors, communications, IoT and AI systems, taking them from prototype to field, and building the engineering teams behind them. The company's probe was originally developed for Scripps Institution of Oceanography. He trained in computer engineering at UC San Diego.

Disclosure

The presenter's company manufactures spectral sensors. This session is educational, not a product presentation. Brand names appear only where a photograph needs a caption, and no pricing is shown.